

# Reading Notes: Analytic Combinatorics

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# Chapter 1

## COMBINATORIAL STRUCTURES AND ORDINARY GENERATING FUNCTIONS

### 1.1 Symbolic enumeration methods

▷ **I.2.** *Unimodal permutations.* A unimodal permutation can be viewed as a sequence of elements increasing to a peak and then decreasing. This structure is represented by:

$$\mathcal{P}^U = \text{SET}(\mathcal{Z}) \star \mathcal{Z}^\blacksquare \star \text{SET}(\mathcal{Z})$$

$$P^U(z) = \int_0^z (e^t \cdot \frac{d}{dt}t \cdot e^t) dt = \frac{1}{2}(e^{2z} - 1)$$

which yields

$$P_n^U = n! [z^n] \frac{1}{2} e^{2z} = n! \frac{1}{2} \frac{2^n}{n!} = 2^{n-1}$$

Alternatively,

$$P_n^U = \sum_{k=1}^n \binom{n-1}{k-1} = \sum_{j=0}^{n-1} \binom{n-1}{j} = 2^{n-1}$$

where  $k$  is the position of  $n$  in the permutation.

### 1.2 Admissible constructions and specifications

### 1.3 Integer compositions and partitions

▷ **I.16.** *Partitions with summands bounded in number and size.* The number of partitions of size  $n$  with at most  $k$  summands, each at most  $l$ , is

$$[z^n] \frac{(1-z)(1-z^2) \cdots (1-z^{k+l})}{((1-z)(1-z^2) \cdots (1-z^k)) \cdot ((1-z)(1-z^2) \cdots (1-z^l))} \quad (1.1)$$

Let  $f(k, l)$  denote the generating function where the coefficient of  $z^n$  represents the number of such partitions. To derive a functional equation, consider the value of the largest

part. There are two mutually exclusive possibilities: the largest part is strictly less than  $l$ , or it is exactly  $l$ . This distinction yields:

$$f(k, l) = f(k, l-1) + z^l f(k-1, l) \quad (1.2)$$

We can derive 1.1 from this relation by introducing  $F_k(x)$ , the generating function for  $f(k, l)$  with respect to the index  $l$ :

$$F_k(x) = \sum_{l=0}^{\infty} f(k, l) x^l$$

For the base case  $k = 0$ , the only valid partition is the empty partition of size 0, implying  $f(0, l) = 1$  for all  $l \geq 0$ . Thus:

$$F_0(x) = \sum_{l=0}^{\infty} 1 \cdot x^l = \frac{1}{1-x}$$

Summing 1.2 over  $l \geq 1$  after multiplying by  $x^l$ , we obtain:

$$\sum_{l=1}^{\infty} f(k, l) x^l - \sum_{l=1}^{\infty} f(k, l-1) x^l = \sum_{l=1}^{\infty} f(k-1, l) (zx)^l$$

- The first term simplifies to  $F_k(x) - 1$ , since  $f(k, 0) = 1$ .
- The second term is simply  $x F_k(x)$ .
- The right-hand side yields  $F_{k-1}(zx) - 1$ .

Plugging these in:

$$(1-x)F_k(x) - 1 = F_{k-1}(zx) - 1 \implies F_k(x) = \frac{1}{1-x} F_{k-1}(zx)$$

Now we apply this functional relation repeatedly:

$$\begin{aligned} F_k(x) &= \frac{1}{1-x} F_{k-1}(zx) \\ &= \frac{1}{1-x} \frac{1}{1-zx} F_{k-2}(z^2x) \\ &= \frac{1}{(1-x)(1-zx) \cdots (1-z^{k-1}x)} F_0(z^k x) \end{aligned}$$

Since  $F_0(x) = \frac{1}{1-x}$ , we have  $F_0(z^k x) = \frac{1}{1-z^k x}$ . Thus:

$$F_k(x) = \sum_{l=0}^{\infty} f(k, l) x^l = \frac{1}{(1-x)(1-zx) \cdots (1-z^k x)}$$

Since all the roots of the denominator are distinct (assuming  $|z| < 1$  or that  $z$  is a formal variable), we can decompose this into a sum of partial fractions:

$$\frac{1}{\prod_{i=0}^k (1-z^i x)} = \sum_{j=0}^k \frac{A_j}{1-z^j x}$$

To find a specific  $A_j$ , we multiply both sides by  $(1 - z^j x)$  and then set  $x = z^{-j}$ :

$$\begin{aligned} A_j &= \left[ \frac{1 - z^j x}{\prod_{i=0}^k (1 - z^i x)} \right]_{x=z^{-j}} \\ &= \frac{1}{\prod_{i=0}^{j-1} (1 - z^i z^{-j}) \prod_{i=j+1}^k (1 - z^i z^{-j})} \\ &= \frac{(-1)^j z^{j(j+1)/2}}{\prod_{i=1}^j (1 - z^i) \prod_{i=1}^{k-j} (1 - z^i)} \end{aligned}$$

The coefficient of  $x^l$  in  $F_k(x)$  is the sum of the coefficients of  $x^l$  in each partial fraction. For a term  $\frac{A_j}{1 - z^j x}$ , the coefficient is  $A_j(z^j)^l$ .

$$\begin{aligned} f(k, l) &= \sum_{j=0}^k A_j (z^j)^l \\ &= \sum_{j=0}^k \frac{(-1)^j z^{j(j+1)/2} \cdot z^{jl}}{\prod_{i=1}^j (1 - z^i) \prod_{i=1}^{k-j} (1 - z^i)} \\ &= \frac{1}{\prod_{i=1}^k (1 - z^i)} \sum_{j=0}^k \binom{k}{j}_z (-1)^j z^{j(j+1)/2 + jl} \end{aligned}$$

where  $\binom{k}{j}_z$  is the Gaussian binomial coefficient:

$$\begin{aligned} \binom{k}{j}_z &= \frac{\prod_{i=1}^k (1 - z^i)}{\prod_{i=1}^j (1 - z^i) \prod_{i=1}^{k-j} (1 - z^i)} \\ &= \frac{(1 - z^k)(1 - z^{k-1}) \cdots (1 - z^{k-j+1})}{(1 - z)(1 - z^2) \cdots (1 - z^j)} \end{aligned}$$

We can rewrite the exponent in the sum:

$$\frac{j(j+1)}{2} + jl = \frac{j(j-1)}{2} + j(l+1)$$

Now the sum becomes

$$f(k, l) = \frac{1}{\prod_{i=1}^k (1 - z^i)} \sum_{j=0}^k \binom{k}{j}_z (-1)^j z^{j(j-1)/2} (z^{l+1})^j$$

The  $q$ -binomial theorem states that for a variable  $X$ :

$$\prod_{i=0}^{k-1} (1 - z^i X) = \sum_{j=0}^k \binom{k}{j}_z (-1)^j z^{j(j-1)/2} X^j \quad (1.3)$$

Plugging  $X = z^{l+1}$  into the theorem yields 1.1:

$$\begin{aligned} f(k, l) &= \frac{(1 - z^{l+1})(1 - z^{l+2}) \cdots (1 - z^{l+k})}{(1 - z^1)(1 - z^2) \cdots (1 - z^k)} \\ &= \frac{(1 - z)(1 - z^2) \cdots (1 - z^{k+l})}{((1 - z)(1 - z^2) \cdots (1 - z^k)) \cdot ((1 - z)(1 - z^2) \cdots (1 - z^l))} \end{aligned}$$

To prove 1.3, let  $P_k(X)$  be the product we are expanding:

$$P_k(X) = \prod_{i=0}^{k-1} (1 - z^i X)$$

We want to show that  $P_k(X) = \sum_{j=0}^k \binom{k}{j}_z (-1)^j z^{j(j-1)/2} X^j$ . Consider the relationship between  $P_k(X)$  and  $P_{k-1}(X)$ . By definition:

$$P_k(X) = (1 - z^{k-1} X) P_{k-1}(X)$$

Now notice what happens if we replace  $X$  with  $zX$  in  $P_k(X)$ :

$$P_k(zX) = \prod_{i=0}^{k-1} (1 - z^{i+1} X) = \frac{1 - z^k X}{1 - X} P_k(X)$$

Rearranging this gives us a functional equation for  $P_k(X)$ :

$$(1 - X) P_k(zX) = (1 - z^k X) P_k(X)$$

Let  $P_k(X) = \sum_{j=0}^k c_j X^j$ . We substitute this power series into the functional equation:

$$(1 - X) \sum_{j=0}^k c_j (zX)^j = (1 - z^k X) \sum_{j=0}^k c_j X^j$$

Now, we compare the coefficients of  $X^j$  on both sides. For a specific power  $j$ :

$$c_j z^j - c_{j-1} z^{j-1} = c_j - c_{j-1} z^k$$

Rearranging to solve for  $c_j$  in terms of  $c_{j-1}$ :

$$c_j = c_{j-1} \frac{z^{j-1} - z^k}{z^j - 1} = -c_{j-1} z^{j-1} \frac{1 - z^{k-j+1}}{1 - z^j}$$

Starting from the base case  $c_0 = P_k(0) = 1$ , we can find  $c_j$  by iteration:

$$c_1 = -c_0 z^0 \frac{1 - z^k}{1 - z^1}, \quad c_2 = -c_1 z^1 \frac{1 - z^{k-1}}{1 - z^2} = (-1)^2 z^{0+1} \frac{(1 - z^k)(1 - z^{k-1})}{(1 - z^1)(1 - z^2)}$$

The exponent of  $z$  becomes  $0 + 1 + 2 + \dots + (j-1) = \frac{j(j-1)}{2}$ . The fraction is the definition of the Gaussian binomial coefficient  $\binom{k}{j}_z$ . Thus, the coefficient is:

$$c_j = (-1)^j z^{j(j-1)/2} \binom{k}{j}_z$$

Plugging the coefficients back into the sum, we obtain:

$$\prod_{i=0}^{k-1} (1 - z^i X) = \sum_{j=0}^k \binom{k}{j}_z (-1)^j z^{j(j-1)/2} X^j$$

This completes the proof.



▷ **I.17.** *Systems of linear Diophantine inequalities.*

$$\{x_1 \geq 0, x_2 \geq 2x_1, x_3 \geq 2x_2, x_4 \geq 2x_3\}$$

We introduce non-negative slack variables  $y_j \in \mathbb{Z}_{\geq 0}$  to quantify the “excess” of each term relative to its lower bound.

$$x_1 = y_1 \tag{1.4}$$

$$x_2 = 2x_1 + y_2 = 2y_1 + y_2 \tag{1.5}$$

$$x_3 = 2x_2 + y_3 = 4y_1 + 2y_2 + y_3 \tag{1.6}$$

$$x_4 = 2x_3 + y_4 = 8y_1 + 4y_2 + 2y_3 + y_4 \tag{1.7}$$

The total weight is  $S = x_1 + x_2 + x_3 + x_4 = 15y_1 + 7y_2 + 3y_3 + y_4$ . Since  $y_j$  are independent and range freely over non-negative integers, the OGF  $F(z)$  is the product of the individual generating functions for each  $y_j$ :

- For  $15y_1$  :  $\sum_{y_1=0}^{\infty} (z^{15})^{y_1} = \frac{1}{1-z^{15}}$
- For  $7y_2$  :  $\sum_{y_2=0}^{\infty} (z^7)^{y_2} = \frac{1}{1-z^7}$
- For  $3y_3$  :  $\sum_{y_3=0}^{\infty} (z^3)^{y_3} = \frac{1}{1-z^3}$
- For  $y_4$  :  $\sum_{y_4=0}^{\infty} (z^1)^{y_4} = \frac{1}{1-z^1}$

Multiplying these together yields the desired OGF:

$$F(z) = \frac{1}{(1-z)(1-z^3)(1-z^7)(1-z^{15})}$$

$$\{x_1 + x_2 \leq x_3\}$$

Following the same approach as in the previous problem, we use slack variables to convert the inequality into an equality of independent parts. The constraint  $x_3 \geq x_1 + x_2$  can be rewritten by introducing a non-negative integer  $y \geq 0$ :

$$x_3 = x_1 + x_2 + y$$

The size of the composition is  $S = x_1 + x_2 + x_3 + x_4 = 2x_1 + 2x_2 + y + x_4$ . Since the variables  $x_1, x_2, y, x_4$  are independent and each ranges freely over the non-negative integers, the OGF is the product of the generating functions for the individual terms in the sum:

- For  $2x_1$  :  $\frac{1}{1-z^2}$
- For  $2x_2$  :  $\frac{1}{1-z^2}$
- For  $y$  :  $\frac{1}{1-z}$
- For  $x_4$  :  $\frac{1}{1-z}$

Multiplying these together gives:

$$F(z) = \frac{1}{(1-z^2)^2(1-z)^2}$$

$$\{x_1 + x_2 \geq x_3\}$$

Since the inequality does not immediately suggest a single slack variable, a more robust approach uses the complement method. Let  $U(z) = (1-z)^{-4}$  be the OGF for four unrestricted variables. The universe of compositions partitions into those where  $x_1 + x_2 < x_3$  and our target class  $x_1 + x_2 \geq x_3$ . We find the OGF for the complement  $x_1 + x_2 < x_3$ . Set  $x_3 = x_1 + x_2 + 1 + y$ , where  $y \geq 0$ . The total sum  $S$  is:

$$S = x_1 + x_2 + (x_1 + x_2 + 1 + y) + x_4 = 2x_1 + 2x_2 + y + x_4 + 1$$

The OGF for this complement, which we denote by  $C(z)$ , is:

$$C(z) = \frac{z}{(1-z^2)^2(1-z)^2}$$

The OGF for our constraint is then:

$$F(z) = U(z) - C(z) = \frac{1}{(1-z)^4} - \frac{z}{(1-z^2)^2(1-z)^2} = \frac{1+z+z^2}{(1-z)^4(1+z)^2}$$

$$\{x_1 + x_2 \leq x_3 + x_4\}$$

To find the OGF for  $\{x_1 + x_2 \leq x_3 + x_4\}$ , we can use the complement method combined with the symmetry of the variables. Let  $A = x_1 + x_2$  and  $B = x_3 + x_4$ . We seek the OGF for compositions satisfying  $A \leq B$ . In the universe of all possible compositions  $(x_1, x_2, x_3, x_4)$ , there are three mutually exclusive cases: 1.  $A < B$ , 2.  $A > B$ , 3.  $A = B$ . By symmetry, the OGF for case 1 must be identical to the OGF for case 2, since  $A$  and  $B$  are built from the same number of variables with the same weights. Denote this common OGF by  $G_{<}(z)$ , and let  $G_{=}(z)$  be the OGF for case 3. The total unrestricted OGF is  $U(z) = \frac{1}{(1-z)^4}$ . Therefore:

$$U(z) = 2G_{<}(z) + G_{=}(z)$$

Our target OGF is  $F(z) = G_{<}(z) + G_{=}(z)$ . Solving the system gives:

$$F(z) = \frac{U(z) + G_{=}(z)}{2}$$

Now  $G_{=}(z)$  is the OGF for compositions in which the sum of the first two variables equals the sum of the last two.

- The number of compositions with  $x_1 + x_2 = n$  is  $[z^n] \frac{1}{(1-z)^2} = n + 1$ .
- The number of compositions with  $x_3 + x_4 = n$  is the same.
- For the equality  $x_1 + x_2 = x_3 + x_4$  to hold, both sums must equal some  $n \geq 0$ .

$$G_{=}(z) = \sum_{n=0}^{\infty} (n+1)^2 z^{2n} = \frac{1+z^2}{(1-z^2)^3}$$

Substituting  $U(z)$  and  $G_=(z)$  into our formula yields:

$$F(z) = \frac{1}{2} \left[ \frac{1}{(1-z)^4} + \frac{1+z^2}{(1-z^2)^3} \right] = \frac{1+z+2z^2}{(1-z)^4(1+z)^3}$$

$$\{x_1 \leq x_2, x_2 \geq x_3, x_3 \leq x_4\}$$

Because the inequalities alternate direction (up, down, up), we cannot use a straightforward chain of slack variable substitutions as we did for  $x_1 \leq x_2 \leq x_3 \leq x_4$ . Instead, we apply the inclusion–exclusion principle to the constraints. We begin with the universe of all unrestricted compositions, whose OGF is  $U(z) = \frac{1}{(1-z)^4}$ . We wish to enforce three conditions:

- $C_1 : x_1 \leq x_2$
- $C_2 : x_2 \geq x_3$
- $C_3 : x_3 \leq x_4$

However, it is simpler to work with chains of “less than or equal” inequalities. Consider the complement of the downward middle step. Take the set of all compositions satisfying both  $x_1 \leq x_2$  and  $x_3 \leq x_4$ . From this set, we subtract those compositions that violate the middle constraint (i.e.,  $x_2 < x_3$ ).

The OGF for the set defined by  $x_1 \leq x_2$  and  $x_3 \leq x_4$ : These two pairs are independent.

- For  $x_1 \leq x_2$ , set  $x_2 = x_1 + y_1$ . The total contribution is  $2x_1 + y_1$ , so the OGF is  $\frac{1}{(1-z^2)(1-z)}$ .
- For  $x_3 \leq x_4$ , the OGF is identically  $\frac{1}{(1-z^2)(1-z)}$ .

Since the two pairs are independent, we multiply their OGFs:

$$A(z) = \frac{1}{(1-z^2)^2(1-z)^2}$$

Now we find the OGF for the violation  $x_1 \leq x_2 < x_3 \leq x_4$  by introducing slack variables:

- $x_1 = y_1$
- $x_2 = y_1 + y_2$
- $x_3 = y_1 + y_2 + 1 + y_3$
- $x_4 = y_1 + y_2 + 1 + y_3 + y_4$

Summing the four expressions gives:

$$S = 4y_1 + 3y_2 + 2y_3 + y_4 + 2$$

The OGF for this violation is therefore:

$$V(z) = \frac{z^2}{(1-z^4)(1-z^3)(1-z^2)(1-z)}$$

The OGF for the zigzag constraint is the OGF for the two independent pairs minus the OGF for the cases where the peak at  $x_2$  fails:

$$\begin{aligned} F(z) &= \frac{1}{(1-z^2)^2(1-z)^2} - \frac{z^2}{(1-z^4)(1-z^3)(1-z^2)(1-z)} \\ &= \frac{1}{(1-z)(1-z^2)} \left[ \frac{1}{(1-z)(1-z^2)} - \frac{z^2}{(1-z^3)(1-z^4)} \right] \end{aligned}$$

▷ **I.18.** *Odd versus distinct summands.* We establish a one-to-one correspondence between partitions into odd parts and partitions into distinct parts.

• Odd  $\rightarrow$  Distinct

1. Group identical odd parts together.
2. If an odd part  $k$  appears  $m$  times, write  $m$  in binary:  $m = 2^{a_1} + 2^{a_2} + \dots$  with distinct powers of two.
3. Replace the  $m$  copies of  $k$  by  $2^{a_1} \cdot k, 2^{a_2} \cdot k, \dots$

• Distinct  $\rightarrow$  Odd

1. Factor each part as  $2^a \times \text{odd}$ .
2. Split it into  $2^a$  copies of that odd factor.
3. Do this for all parts; now all parts are odd.
4. Sort the resulting parts to obtain a partition into odd parts.

As an example:

$$\begin{aligned} 3 + 3 + 3 + 3 + 3 + 1 + 1 &= 5 \times 3 + 2 \times 1 \\ &= (2^2 + 2^0) \times 3 + 2^1 \times 1 \\ &= 4 \times 3 + 1 \times 3 + 2 \times 1 \\ &= 12 + 3 + 2 \end{aligned}$$

$$\begin{aligned} 12 + 3 + 2 &= 4 \times 3 + 3 + 2 \times 1 \\ &= (3 + 3 + 3 + 3) + 3 + (1 + 1) \\ &= 3 + 3 + 3 + 3 + 3 + 1 + 1 \end{aligned}$$

▷ **I.19.** *Euler's pentagonal number theorem.* To find a recurrence for the partition numbers, we consider the reciprocal of the generating function  $P(z)$ :

$$A(z) = \prod_{n=1}^{\infty} (1 - z^n)$$

Euler's Pentagonal Number Theorem states that this product expands into the following sparse power series:

$$\begin{aligned} A(z) &= \sum_{k=-\infty}^{\infty} (-1)^k z^{k(3k+1)/2} \\ &= \sum_{k=-\infty}^{\infty} (-1)^k z^{k(3k-1)/2} \\ &= 1 - z^1 - z^2 + z^5 + z^7 - z^{12} - z^{15} + \dots \end{aligned}$$

The exponents  $g_k = \frac{k(3k-1)}{2}$  are the generalized pentagonal numbers. The product of  $P(z)$  and  $A(z)$  must equal 1:

$$\sum_{n=0}^{\infty} p(n)z^n \cdot \sum_{k=-\infty}^{\infty} (-1)^k z^{g_k} = 1$$

Comparing the coefficient of  $z^n$  on both sides yields the algorithmically useful recurrence:

$$p(n) = p(n-1) + p(n-2) - p(n-5) - p(n-7) + p(n-12) + p(n-15) \cdots$$

A Python implementation is provided below:

```

1 def compute_partitions(N):
2     P = [0] * (N + 1)
3     P[0] = 1 # Base case: p(0) = 1
4
5     for i in range(1, N + 1):
6         k = 1
7         while True:
8             # Calculate pentagonal numbers for k and -k
9             g_k_pos = k * (3 * k - 1) // 2
10            g_k_neg = k * (3 * k + 1) // 2
11
12            # Determine sign: +1 for k=1,2 (terms 1,2), -1 for k
13            # =3,4 (terms 3,4)...
14            # Simplest way: (-1)^(k-1)
15            sign = 1 if k % 2 == 1 else -1
16
17            # Add p(i - g_k_pos)
18            if g_k_pos <= i:
19                P[i] += sign * P[i - g_k_pos]
20            else:
21                break # No more terms will fit
22
23            # Add p(i - g_k_neg)
24            if g_k_neg <= i:
25                P[i] += sign * P[i - g_k_neg]
26            else:
27                break
28
29            k += 1
30
31     return P

```

▷ **I.20.** *A digital surprise.*

$$\begin{aligned}\varphi &= (1 - 10^{-1})(1 - 10^{-2})(1 - 10^{-3})(1 - 10^{-4}) \dots \\ &= \left[ \prod_{n=1}^{\infty} (1 - z^n) \right]_{z=.1} \\ &= [1 - z^1 - z^2 + z^5 + z^7 - z^{12} - z^{15} + z^{22} + z^{26} - \dots]_{z=.1} \\ &= .9 - .1 - .01 + .00001 + .0000001 - .000000000001 - \dots\end{aligned}$$

The partial sums are:

- $\varphi_1 = .999999999999999999999999$
- $\varphi_2 = .899999999999999999999999$
- $\varphi_3 = .889999999999999999999999$
- $\varphi_4 = .890009999999999999999999$
- $\varphi_5 = .890010099999999999999999$

- $\varphi_6 = .8900100999989999999999999999$
- $\varphi_7 = .8900100999989989999999999999$
- $\varphi_8 = .89001009999899900000000999$

▷ **I.21. Lattice points.** We are interested in the number of integer solutions to the inequality  $\sum_{i=1}^d x_i^2 \leq n^2$ . First, consider a single coordinate  $x_i \in \mathbb{Z}$ . The “size” contributed by this coordinate to the sum of squares is  $x_i^2$ . The generating function for the possible values of a single coordinate, where the exponent of  $z$  represents the square of the coordinate, is:

$$\Theta(z) = \sum_{x \in \mathbb{Z}} z^{x^2} = 1 + 2 \sum_{n=1}^{\infty} z^{n^2}$$

Since each coordinate  $x_1, \dots, x_d$  is chosen independently, the generating function for the sum of squares  $\sum x_i^2$  is the  $d$ -th power of the individual generating function:

$$S(z) = (\Theta(z))^d = \sum_{k=0}^{\infty} r_d(k) z^k$$

In this series, the coefficient  $[z^k]S(z)$  (denoted as  $r_d(k)$ ) is exactly the number of ways to represent the integer  $k$  as a sum of  $d$  squares. Geometrically, this is the number of lattice points on the sphere (the boundary) of radius  $\sqrt{k}$ . The number of points  $N(n, d)$  in the closed ball of radius  $n$  is:

$$N(n, d) = \sum_{k=0}^{n^2} r_d(k)$$

In generating functions, the prefix sum of a sequence’s coefficients is obtained by multiplying the generating function by  $\frac{1}{1-z}$ . Therefore:

$$\sum_{k=0}^{\infty} \left( \sum_{j=0}^k r_d(j) \right) z^k = \frac{1}{1-z} S(z) = \frac{1}{1-z} (\Theta(z))^d$$

To find the total count up to  $k = n^2$ , we simply extract the coefficient of  $z^{n^2}$ :

$$N(n, d) = [z^{n^2}] \frac{1}{1-z} (\Theta(z))^d$$

## 1.4 Words and regular languages

▷ **I.23. Alice, Bob, and coding bounds.** Following Proposition I.4, the class  $\mathcal{C}$  of binary strings avoiding the pattern 11 has the OGF:

$$C(z) = \frac{1+z}{1-z-z^2}$$

To account for a variable-length coding scheme where codewords are selected from  $\bigcup_{j=0}^L \mathcal{C}_j$ , we consider the sum of the first  $L+1$  coefficients. This is represented by the OGF:

$$S(z) = \frac{C(z)}{1-z} = \frac{1+z}{(1-z)(1-z-z^2)}$$

The asymptotic growth of the sum  $\sum_{j=0}^L C_j$  is governed by the smallest singularity of  $S(z)$ . Since the pole at  $z = 1/\varphi$  (where  $\varphi = \frac{1+\sqrt{5}}{2}$ ) is closer to the origin than the pole at  $z = 1$ , the exponential growth is dominated by  $C(z)$ . Specifically, as  $L \rightarrow \infty$ :

$$\sum_{j=0}^L C_j \sim A \cdot \varphi^L$$

for some constant  $A$ . To find the required code length  $L(n)$  for  $n$  bits of information, we solve for  $2^n \leq \sum_{j=0}^L C_j$  by taking the base-2 logarithm:

$$n \leq \log_2 \left( \sum_{j=0}^L C_j \right) = L \log_2 \varphi + O(1)$$

Rearranging for  $L$ , we obtain the lower bound:

$$L \geq \frac{n}{\log_2 \varphi} + O(1) \approx 1.440420n + O(1)$$

▷ **I.27. Congruence languages.** The DFA  $\mathcal{A}$  for binary strings congruent to 2 (mod 7) is structured as follows:

- **States:** A set  $\mathcal{Q} = \{S_0, \dots, S_6\}$ , where each state  $S_r$  represents the current remainder  $r$  modulo 7.
- **Initial State:**  $S_0$  (representing the value 0).
- **Accepting State:**  $\overline{\mathcal{Q}} = \{S_2\}$ .
- **Alphabet:**  $\mathcal{A} = \{0, 1\}$ .

Each transition follows the rule  $q_r \circ b = q_{(2r+b) \pmod{7}}$ :

Table 1.1: State Transition

| Current State (Remainder) | Input 0 ( $2r \pmod{7}$ ) | Input 1 ( $2r + 1 \pmod{7}$ ) |
|---------------------------|---------------------------|-------------------------------|
| $S_0$                     | $S_0$                     | $S_1$                         |
| $S_1$                     | $S_2$                     | $S_3$                         |
| $S_2$                     | $S_4$                     | $S_5$                         |
| $S_3$                     | $S_6$                     | $S_0$                         |
| $S_4$                     | $S_1$                     | $S_2$                         |
| $S_5$                     | $S_3$                     | $S_4$                         |
| $S_6$                     | $S_5$                     | $S_6$                         |

▷ **I.29. The forward equations.** By a “last-letter decomposition”, the set  $\mathcal{M}_k$  of words leading to state  $q_k$  from the initial state  $q_0$  satisfies the combinatorial relation:

$$\mathcal{M}_k \cong \llbracket k = 0 \rrbracket \cdot \{\epsilon\} + \sum_{j, \alpha: q_j \circ \alpha = q_k} (\mathcal{M}_j \times \{\alpha\})$$



where  $\llbracket \cdot \rrbracket$  is the Iverson bracket. This translates to a linear system of OGFs:

$$M_k(z) = \delta_{k,0} + z \sum_j M_j(z) T_{j,k}$$

In vector notation, with  $\mathbf{M}(z)$  as a column vector and  $\mathbf{e}_0$  the first basis vector, the system is a transposed version of the backward equations:

$$\mathbf{M}(z)^T = \mathbf{e}_0^T + z \mathbf{M}(z)^T T$$

▷ **I.33.** *Waiting times in strings.* To illustrate the formulas, let's define a prefix language  $\hat{\mathcal{L}}$  containing one word of length 2 and one word of length 3. Let  $\hat{\mathcal{L}} = \{aa, bbb\}$ .

1. Verification of Prefix Property: The word  $aa$  is not a prefix of  $bbb$ , and  $bbb$  is not a prefix of  $aa$ . Since no word is a strict prefix of another, this is a valid prefix language  $\hat{\mathcal{L}}$ .
2. The Generating Function  $\hat{L}(z)$ : The OGF is the sum of  $z$  raised to the power of each word's length:  $\hat{L}(z) = z^2 + z^3$ .
3. Probability:  $\hat{L}(1/2)$ : This represents the probability that a random infinite string starts with a word in  $\mathcal{L}$ :  $\hat{L}(1/2) = (1/2)^2 + (1/2)^3 = 1/4 + 1/8 = 3/8$ . This means there is a 37.5% chance that a random string begins with a word from this language.
4. Expected Time:  $\frac{1}{2} \hat{L}'(1/2)$ . To find the expected time (or more precisely, the expected “matched length”) at which a word in  $\mathcal{L}$  is first encountered:  $\hat{L}'(z) = 2z + 3z^2 \implies \hat{L}'(1/2) = 1.75 \implies E = 0.875$ .
5. Breakdown of the Result: The total expected value of 0.875 is formed as follows:
  - Word “aa”: Probability 1/4 of length 2  $\rightarrow$  Contribution:  $2 \times 0.25 = 0.5$ .
  - Word “bbb”: Probability 1/8 of length 3  $\rightarrow$  Contribution:  $3 \times 0.125 = 0.375$ .
  - No Match: Probability 5/8 of length 0  $\rightarrow$  Contribution:  $0 \times 0.625 = 0$ .

▷ **I.34.** *A probabilistic paradox on strings.* Consider a binary pattern of length  $k$ . The generating function  $T(z)$  for the first occurrence of this pattern is given by:

$$T(z) = \frac{z^k}{z^k + (1 - mz)c(z)} = \frac{z^k}{z^k + (1 - 2z)c(z)}$$

The average waiting time (or expected occurrence time) can be derived directly from  $T(z)$  by evaluating its derivative at the point of the alphabet's inverse size ( $z = 1/2$ ):

$$\left[ z \frac{d}{dz} T(z) \right]_{z=1/2} = 2^k c(1/2)$$

## 1.5 Tree structures

▷ **I.39.** *Unary–binary trees and Motzkin numbers.* We identify the characteristic polynomial as  $\phi(u) = 1 + u + u^2$ . To find the number of trees with  $n$  nodes,  $M_n = [z^n]M(z)$ , we apply the Lagrange Inversion Theorem:

$$\begin{aligned}
 M_n &= \frac{1}{n} [u^{n-1}] \phi(u)^n \\
 &= \frac{1}{n} [u^{n-1}] (1 + u + u^2)^n \\
 &= \frac{1}{n} [u^{n-1}] \sum_j \binom{n}{j} (u + u^2)^j \\
 &= \frac{1}{n} \sum_j \binom{n}{j} [u^{n-1}] u^j (1 + u)^j \\
 &= \frac{1}{n} \sum_j \binom{n}{j} [u^{n-j-1}] \sum_l \binom{j}{l} u^l \\
 &= \frac{1}{n} \sum_j \binom{n}{j} \binom{j}{n-j-1} \\
 &= \frac{1}{n} \sum_k \binom{n}{n-k} \binom{n-k}{n-(n-k)-1} \\
 &= \frac{1}{n} \sum_k \binom{n}{k} \binom{n-k}{k-1}
 \end{aligned}$$

▷ **I.41.** *The Lagrangean form of Schröder's GF.* We want to prove the last equality

$$\frac{(-1)^{n-1}}{n} \sum_k (-2)^k \binom{n}{k+1} \binom{n+k-1}{k} = \frac{1}{n} \sum_{k=0}^{n-2} \binom{2n-k-2}{n-1} \binom{n-2}{k}$$

Multiplying by  $n$ , the statement is equivalent to

$$(-1)^{n-1} \sum_{k=0}^{n-1} (-2)^k \binom{n}{k+1} \binom{n+k-1}{k} = \sum_{k=0}^{n-2} \binom{2n-k-2}{n-1} \binom{n-2}{k}$$

**Step 1. Re-index the left sum.** Using  $\binom{n}{k+1} = \binom{n}{n-1-k}$  and  $\binom{n+k-1}{k} = \binom{n+k-1}{n-1}$ , set  $j = n-1-k$ :

$$L = (-1)^{n-1} \sum_{j=0}^{n-1} (-2)^{n-1-j} \binom{n}{j} \binom{2n-2-j}{n-1} = 2^{n-1} \sum_{j=0}^{n-1} \left(-\frac{1}{2}\right)^j \binom{n}{j} \binom{2n-2-j}{n-1}$$

**Step 2. Express as a coefficient extraction.** Since  $\binom{2n-2-j}{n-1} = [z^{n-1}](1+z)^{2n-2-j}$ ,

we have

$$\begin{aligned}
L &= 2^{n-1} \sum_{j=0}^{n-1} \left(-\frac{1}{2}\right)^j \binom{n}{j} [z^{n-1}](1+z)^{2n-2-j} \\
&= 2^{n-1} [z^{n-1}] \sum_{j=0}^{n-1} \left(-\frac{1}{2}\right)^j \binom{n}{j} (1+z)^{2n-2-j} \\
&= 2^{n-1} [z^{n-1}](1+z)^{2n-2} \sum_{j=0}^{n-1} \binom{n}{j} \left(-\frac{1}{2(1+z)}\right)^j \\
&= 2^{n-1} [z^{n-1}](1+z)^{2n-2} \left(1 - \frac{1}{2(1+z)}\right)^n \\
&= 2^{n-1} [z^{n-1}](1+z)^{2n-2} \frac{(1+2z)^n}{2^n(1+z)^n} \\
&= \frac{1}{2} [z^{n-1}](1+z)^{n-2}(1+2z)^n
\end{aligned}$$

**Step 3. Reverse the polynomial.** Let  $P(z) = (1+z)^{n-2}(1+2z)^n$ , of degree  $2n-2$ . The coefficient of  $z^{n-1}$  in  $P(z)$  equals the coefficient of  $z^{n-1}$  in its reversed polynomial  $z^{2n-2}P(1/z)$ :

$$z^{2n-2}P(1/z) = z^{2n-2} \left(1 + \frac{1}{z}\right)^{n-2} \left(1 + \frac{2}{z}\right)^n = (z+1)^{n-2}(z+2)^n$$

Hence

$$L = \frac{1}{2} [z^{n-1}](1+z)^{n-2}(2+z)^n$$

**Step 4. Express the right sum as a coefficient.** Similarly,  $\binom{2n-k-2}{n-1} = [z^{n-1}](1+z)^{2n-k-2}$ , so

$$\begin{aligned}
R &= \sum_{k=0}^{n-2} \binom{2n-k-2}{n-1} \binom{n-2}{k} \\
&= \sum_{k=0}^{n-2} [z^{n-1}](1+z)^{2n-k-2} \binom{n-2}{k} \\
&= [z^{n-1}] \sum_{k=0}^{n-2} (1+z)^{2n-k-2} \binom{n-2}{k} \\
&= [z^{n-1}](1+z)^{2n-2} \sum_{k=0}^{n-2} \binom{n-2}{k} (1+z)^{-k} \\
&= [z^{n-1}](1+z)^{2n-2} \left(1 + \frac{1}{1+z}\right)^{n-2} \\
&= [z^{n-1}](1+z)^{2n-2} \frac{(2+z)^{n-2}}{(1+z)^{n-2}} \\
&= [z^{n-1}](1+z)^n(2+z)^{n-2}
\end{aligned}$$

**Step 5. Show equality of the two coefficient expressions.** We need to show  $\frac{1}{2}(1+z)^{n-2}(2+z)^n$  and  $(1+z)^n(2+z)^{n-2}$  have the same coefficient of  $z^{n-1}$ . Their

difference is

$$\begin{aligned}\Delta &= \frac{1}{2}[z^{n-1}](1+z)^{n-2}(2+z)^n - [z^{n-1}](1+z)^n(2+z)^{n-2} \\ &= \frac{1}{2}[z^{n-1}](1+z)^{n-2}(2+z)^{n-2}[(2+z)^2 - 2(1+z)^2] \\ &= \frac{1}{2}[z^{n-1}](1+z)^{n-2}(2+z)^{n-2}(2-z^2)\end{aligned}$$

We must prove  $[z^{n-1}]f(z) = 0$ , where

$$f(z) = (1+z)^{n-2}(2+z)^{n-2}(2-z^2)$$

**Step 6. A symmetry argument.**  $f(z)$  is a polynomial of degree  $2n-2$ . Compute  $f(2/z)$ :

$$\begin{aligned}f(2/z) &= (1+2/z)^{n-2}(2+2/z)^{n-2}(2-(2/z)^2) \\ &= \left(\frac{2+z}{z}\right)^{n-2} \left(\frac{2(1+z)}{z}\right)^{n-2} \left(2 - \frac{4}{z^2}\right) \\ &= \frac{(2+z)^{n-2}(1+z)^{n-2}2^{n-2}}{z^{2n-4}} \cdot \frac{-2(2-z^2)}{z^2} \\ &= -2^{n-1} \frac{f(z)}{z^{2n-2}}\end{aligned}$$

Hence  $z^{2n-2}f(2/z) = -2^{n-1}f(z)$ . Writing  $f(z) = \sum_{k=0}^{2n-2} a_k z^k$ , the coefficient of  $z^{n-1}$  corresponds to  $k = n-1$ . Equating coefficients gives  $a_{n-1}2^{n-1} = -2^{n-1}a_{n-1} \implies a_{n-1} = 0$ .

▷ **I.42.** *Faster determination of Schröder numbers.*

$$S(z) = \frac{1}{4}(1+z-\sqrt{1-6z+z^2})$$

Consider in general

$$p(z)S(z) + q(z) = \sqrt[k]{r(z)}$$

where  $p(z)$ ,  $q(z)$ , and  $r(z)$  are functions of  $z$ . Taking the derivative with respect to  $z$ ,

$$p'(z)S(z) + p(z)S'(z) + q'(z) = \frac{1}{k}r(z)^{\frac{1}{k}-1}r'(z)$$

Multiplying both sides by  $r(z)$ ,

$$p'(z)r(z)S(z) + p(z)r(z)S'(z) + q'(z)r(z) = \frac{1}{k}(p(z)S(z) + q(z))r'(z)$$

Rearranging,

$$p(z)r(z)S'(z) + (p'(z)r(z) - \frac{1}{k}p(z)r'(z))S(z) = \frac{1}{k}q(z)r'(z) - q'(z)r(z)$$

Plugging  $p(z) = -4$ ,  $q(z) = 1+z$ ,  $k = 2$ , and  $r(z) = 1-6z+z^2$  into the equation and simplifying yields

$$(z^2 - 6z + 1)S'(z) + (3-z)S(z) = 1-z$$

from which the recurrence follows.

▷ **I.43.** *Fast determination of the Cayley–Pólya numbers.*

$$H(z) = z \exp \left( H(z) + \frac{1}{2}H(z^2) + \cdots \right)$$

**Step 1. Take the Natural Logarithm.** First, we take the log of both sides to bring down the terms in the exponent:

$$\ln H(z) = \ln z + \sum_{k=1}^{\infty} \frac{1}{k} H(z^k)$$

**Step 2. Differentiate with respect to  $z$ .** Now, we apply the derivative operator  $\frac{d}{dz}$ :

$$\frac{H'(z)}{H(z)} = \frac{1}{z} + \sum_{k=1}^{\infty} \frac{1}{k} H'(z^k) \cdot k z^{k-1} = \frac{1}{z} + \sum_{k=1}^{\infty} z^{k-1} H'(z^k)$$

**Step 3. Multiply by  $zH(z)$ .** To clear the fractions and align the powers of  $z$ , we multiply the entire equation by  $zH(z)$ :

$$zH'(z) = H(z) \left( 1 + \sum_{k=1}^{\infty} z^k H'(z^k) \right)$$

**Step 4. Substitute Power Series.** We represent  $H(z)$  as a formal power series  $\sum_{n=1}^{\infty} H_n z^n$ . Note that:

- $zH'(z) = \sum_{n=1}^{\infty} n H_n z^n$
- $z^k H'(z^k) = \sum_{m=1}^{\infty} m H_m z^{mk}$

Plugging these into our equation:

$$\sum_{n=1}^{\infty} n H_n z^n = H(z) + H(z) \sum_{k=1}^{\infty} \sum_{m=1}^{\infty} m H_m z^{mk}$$

**Step 5. The Recurrence Relation.** By extracting the coefficient of  $z^n$  from both sides, we arrive at the recurrence. The double sum represents a convolution where we look for indices such that  $mk + j = n$  (where  $j$  is the index from the  $H(z)$  term). The standard form for the recurrence of these coefficients is:

$$(n-1)H_n = \sum_{l=1}^{n-1} H_{n-l} \left( \sum_{d|l} d H_d \right)$$

Or, more commonly expressed by isolating  $H_n$ :

$$H_n = \frac{1}{n-1} \sum_{l=1}^{n-1} H_{n-l} \left( \sum_{d|l} d H_d \right) = \frac{1}{n-1} \sum_{l=1}^{n-1} a_l H_{n-l}$$

To manually calculate the first few values using this recurrence, we start with the base case  $H_1 = 1$  representing the root of the tree.

- To find  $H_2$ , we set  $n = 2$ . The summation runs for  $l = 1$ . The only divisor of 1 is 1.  $a_1 = \sum_{d|1} dH_d = 1 \cdot H_1 = 1 \implies H_2 = \frac{1}{2-1}(H_{2-1} \cdot a_1) = 1$ .
- For  $H_3$ , the summation runs for  $l = 1$  and  $l = 2$ . The divisors of 2 are 1 and 2.  $a_2 = \sum_{d|2} dH_d = 1 \cdot H_1 + 2 \cdot H_2 = 3 \implies H_3 = \frac{1}{3-1}(H_{3-1} \cdot a_1 + H_{3-2} \cdot a_2) = 2$ .
- For  $H_4$ , the summation runs for  $l = 1, 2, 3$ . The divisors of 3 are 1 and 3.  $a_3 = \sum_{d|3} dH_d = 1 \cdot H_1 + 3 \cdot H_3 = 7 \implies H_4 = \frac{1}{4-1}(H_{4-1} \cdot a_1 + H_{4-2} \cdot a_2 + H_{4-3} \cdot a_3) = 4$ .

▷ **I.50. Excursions, bridges, and meanders.** A meander of final altitude  $k$  consists of  $k$  "up-steps" ( $\mathcal{Z}$ ) that lead to new "floors", with Dyck paths (excursions  $\mathcal{D}$ ) occurring between them and at the very end.

$$\mathcal{M} = \mathcal{D} + (\mathcal{D} \times \mathcal{Z} \times \mathcal{D}) + (\mathcal{D} \times \mathcal{Z} \times \mathcal{D} \times \mathcal{Z} \times \mathcal{D}) + \cdots = \mathcal{D} \times \sum_{k=0}^{\infty} (\mathcal{Z} \times \mathcal{D})^k$$

A bridge is a sequence of primitive excursions. A primitive excursion starts at 0, takes one step (up or down), wanders without returning to 0, and then takes one step back to 0.

- **Going Above:** Starts with +1, ends with -1, with a Dyck path in between:  $\mathcal{Z} \times \mathcal{D} \times \mathcal{Z}$
- **Going Down:** Starts with -1, ends with 1, with mirrored a Dyck path in between:  $\mathcal{Z} \times \mathcal{D} \times \mathcal{Z}$

$$\mathcal{B} = \text{SEQ}(\mathcal{Z} \times \mathcal{D} \times \mathcal{Z} + \mathcal{Z} \times \mathcal{D} \times \mathcal{Z}) = \text{SEQ}(2 \times \mathcal{Z}^2 \times \mathcal{D})$$

We want to show  $M_n = \binom{n}{\lfloor n/2 \rfloor}$ . A meander represents a path that stays in the first quadrant. Let  $N(n, k)$  be the number of paths of length  $n$  ending at altitude  $k$ . By the reflection principle, the number of paths that hit  $-1$  and end at  $k \geq 0$  is exactly the same as the number of unrestricted paths that end at  $-k-2 < -1$ . Thus, Meanders end at  $k = (\text{Paths that end at } k) - (\text{Paths that hit } -1 \text{ and end at } k) = (\text{Paths that end at } k) - (\text{Paths end at } -k-2)$ . Summing over all possible final altitudes  $k \geq 0$ , we get a telescoping sum:

$$M_n = \sum_{k \geq 0} [N(n, k) - N(n, k+2)]$$

- $n$  is even ( $n = 2m$ ):  $M_n = N(n, 0) = N(2m, 0) = \binom{2m}{\frac{2m+0}{2}} = \binom{2m}{m} = \binom{n}{\lfloor n/2 \rfloor}$
- $n$  is odd ( $n = 2m+1$ ):  $M_n = N(n, 1) = N(2m+1, 1) = \binom{2m+1}{\frac{2m+1+1}{2}} = \binom{2m+1}{m} = \binom{n}{\lfloor n/2 \rfloor}$

▷ **I.17. The complexity of boolean functions.**

$$D_n = O(16^n m^n n^{-3/2}) \implies \sum_{j=0}^v D_j = O(16^v m^v v^{-3/2})$$

We prove the following general statement.

Let  $c > 0$  and  $\alpha > 0$ . Define  $a_n = c^n n^{-\alpha}$  for  $n \geq 1$  and let  $a_0 > 0$  be a constant. If  $D_n = O(a_n)$ , then  $\sum_{j=0}^v D_j = O(c^v v^{-\alpha})$  as  $v \rightarrow \infty$ .

Proof. Because  $D_n = O(a_n)$ , there exists  $M > 0$  and  $N \geq 1$  such that  $D_n \leq Mc^n n^{-\alpha}$  for all  $n \geq N$ . For  $v \geq N$  we split

$$\sum_{j=0}^v D_j \leq \underbrace{\sum_{j=0}^{N-1} D_j}_{C_0} + M \sum_{j=N}^v c^j j^{-\alpha}$$

Focus on  $S(v) = \sum_{j=N}^v c^j j^{-\alpha}$ . Split at  $j = \lfloor v/2 \rfloor$ :

$$S(v) = \underbrace{\sum_{j=N}^{\lfloor v/2 \rfloor} c^j j^{-\alpha}}_{\text{early}} + \underbrace{\sum_{j=\lfloor v/2 \rfloor + 1}^v c^j j^{-\alpha}}_{\text{tail}}$$

**Early terms.** For  $j \leq v/2$  we have  $c^j \leq c^{v/2}$ . Also since  $\alpha > 0$ , the factor  $j^{-\alpha}$  is decreasing; its maximum is at  $j = N$ , so  $j^{-\alpha} \leq N^{-\alpha}$ . Thus

$$\sum_{j=N}^{\lfloor v/2 \rfloor} c^j j^{-\alpha} \leq N^{-\alpha} \sum_{j=N}^{\lfloor v/2 \rfloor} c^j \leq N^{-\alpha} \frac{c^{v/2+1}}{c-1} = O(c^{v/2})$$

Because  $c > 1$ ,  $c^{v/2}$  is exponentially smaller than  $c^v$ , so this part is  $o(c^v v^{-\alpha})$ .

**Tail terms.** For  $j \geq v/2$ ,  $j^{-\alpha} \leq (v/2)^{-\alpha} = 2^\alpha v^{-\alpha}$ . Consequently

$$\sum_{j=\lfloor v/2 \rfloor + 1}^v c^j j^{-\alpha} \leq 2^\alpha v^{-\alpha} \sum_{j=\lfloor v/2 \rfloor + 1}^v c^j \leq 2^\alpha v^{-\alpha} \frac{c^{v+1}}{c-1} = O(c^v v^{-\alpha})$$

Putting it together,  $S(v) = o(c^v v^{-\alpha}) + O(c^v v^{-\alpha}) = O(c^v v^{-\alpha})$ . Hence

$$\sum_{j=0}^v D_j = C_0 + MS(v) = O(1) + O(c^v v^{-\alpha}) = O(c^v v^{-\alpha})$$

Now we show

$$\sum_{j=0}^v D_j = O(16^v m^v v^{-3/2}) = o(\|\Omega\|)$$

by taking for  $v$  the value

$$v = \frac{2^m}{4 + \log_2 m}$$

Note that

$$v \log_2 16m = \frac{2^m}{4 + \log_2 m} (4 + \log_2 m) = 2^m \implies 16^v m^v = 2^{2^m}$$

Hence the whole bound becomes

$$\sum_{j=0}^v D_j = O(2^{2^m} v^{-3/2})$$

Dividing by  $\|\Omega\|$ ,

$$\frac{\sum_{j=0}^v D_j}{\|\Omega\|} = O(v^{-3/2})$$

As  $m \rightarrow \infty$ ,  $v \rightarrow \infty$ , so  $v^{-3/2} \rightarrow 0$ ; therefore

$$\frac{\sum_{j=0}^v D_j}{\|\Omega\|} \rightarrow 0$$

which is exactly  $\sum_{j=0}^v D_j = o(\|\Omega\|)$ .

## 1.6 Additional constructions

▷ **I.62.** *Sequences without repeated components.* A sequence without repeated components is essentially an ordered arrangement of a subset of distinct elements from a base class  $\mathcal{B}$ . To derive its generating function, we first consider the construction of a set of distinct elements (the power set), then account for the ordering. The generating function for the power set construction  $\text{PSET}(\mathcal{B})$  is given by:

$$A(z, u) = \prod_{\beta \in \mathcal{B}} (1 + uz^{|\beta|}) = \exp \left( \sum_{j \geq 1} (-1)^{j-1} \frac{u^j}{j} B(z^j) \right)$$

Here, the coefficient  $[u^k]A(z, u) = f^{(k)}(z)$  represents the generating function for sets containing exactly  $k$  distinct components. While the power set construction satisfies the "no-repeat" constraint, it treats the collection as unordered. To transform these sets into sequences, we multiply the generating function for sets of size  $k$  by  $k!$  to account for all possible permutations. Using the Eulerian integral representation  $k! = \int_0^\infty e^{-u} u^k du$ , we can express the generating function for distinct sequences,  $\mathcal{S}_{\text{distinct}}(z)$ , as follows:

$$\begin{aligned} \mathcal{S}_{\text{distinct}}(z) &= \sum_{k \geq 0} (\text{Sets of size } k) \cdot k! \\ &= \sum_{k \geq 0} f^{(k)}(z) \cdot k! \\ &= \sum_{k \geq 0} f^{(k)}(z) \cdot \int_0^\infty e^{-u} u^k du \\ &= \sum_{k \geq 0} \int_0^\infty f^{(k)}(z) e^{-u} u^k du \\ &= \int_0^\infty \sum_{k \geq 0} f^{(k)}(z) e^{-u} u^k du \\ &= \int_0^\infty \sum_{k \geq 0} f^{(k)}(z) u^k \cdot e^{-u} du \\ &= \int_0^\infty A(z, u) \cdot e^{-u} du \\ &= \int_0^\infty \exp \left( \sum_{j \geq 1} (-1)^{j-1} \frac{u^j}{j} B(z^j) \right) e^{-u} du \end{aligned}$$

## 1.7 Perspective



## Chapter 2

# LABELLED STRUCTURES AND EXPONENTIAL GENERATING FUNCTIONS

### 2.1 Labelled classes

### 2.2 Admissible labelled constructions

### 2.3 Surjections, set partitions, and words

### 2.4 Alignments, permutations, and related structures

### 2.5 Labelled trees, mappings, and graphs

▷ II.19. *Labelled hierarchies.*

$$L = z + e^L - 1 - L \implies L(z) = T\left(\frac{1}{2}e^{z/2-1/2}\right) + \frac{z}{2} - \frac{1}{2}$$

**Step 1. Re-arranging the target equation.** We start with the given equation, moving all terms involving  $L$  to the left side and constant/independent terms to the right:

$$2L + 1 - e^L = z$$

This form does not immediately match  $T(z) = ze^{T(z)}$ . To make them compatible, we should attempt to isolate an exponential term that mirrors the functional form of the tree function.

**Step 2. Utilizing the Lambert W identity.** The Cayley tree function  $T(z)$  is related to the **Lambert W function** (specifically  $W(-z)$ ). The identity  $T(z) = ze^{T(z)}$  can be rewritten as:

$$-T(z)e^{-T(z)} = -z$$

Which implies  $T(z) = -W(-z)$ .

**Step 3. Substitution and Solving.** Let's rearrange the original equation to isolate the exponential:

$$e^L = 2L - z + 1$$

For equations where a variable appears both linearly and in the exponent, the solution is typically logarithmic or involves the  $W$  function. Let's rewrite the equation for  $L$  in the form  $f(L)e^{f(L)} = \text{something}$ :

$$1 = (2L - z + 1)e^{-L}$$

Now, we want the term inside the parentheses to match the exponent. Let  $u$  be the linear part related to  $L$ . If we aim for  $u = \frac{z-1}{2} - L$ , then:

$$2u = z - 1 - 2L$$

Substituting this back into our equation:

$$1 = (-2u)e^{-L}$$

Since  $u = \frac{z-1}{2} - L$ , we have  $-L = u - \frac{z-1}{2}$ . Substituting this into the exponent:

$$1 = -2u \cdot e^{u - \frac{z-1}{2}} = -2u \cdot e^u \cdot e^{\frac{1-z}{2}}$$

Now, isolate  $ue^u$ :

$$ue^u = -\frac{1}{2}e^{\frac{z-1}{2}}$$

Since  $ue^u = v$  implies  $u = W(v)$ , and we know  $W(v) = -T(-v)$ :

$$u = W\left(-\frac{1}{2}e^{\frac{z-1}{2}}\right) = -T\left(\frac{1}{2}e^{\frac{z-1}{2}}\right)$$

Substituting  $u$  back in:

$$\frac{z-1}{2} - L = -T\left(\frac{1}{2}e^{\frac{z-1}{2}}\right)$$

Solving for  $L$ :

$$L = T\left(\frac{1}{2}e^{\frac{z-1}{2}}\right) + \frac{z-1}{2}$$

**II.5.2. Mappings and functional graphs.** Let  $\mathcal{T}$  be the class of all Cayley trees. Then

$$F = (1 - T)^{-1} \implies F_n = n^n$$

This result follows from Lagrange inversion. The standard formulation states that

$$z = T/\phi(T) \implies [z^n]T(z) = \frac{1}{n}[w^{n-1}]\phi(w)^n$$

More generally,

$$[z^n]H(T(z)) = \frac{1}{n}[w^{n-1}]H'(w)\phi(w)^n$$

Substituting  $H(w) = (1 - w)^{-1}$  and  $\phi(w) = e^w$  yields

$$\begin{aligned}
F_n &= n![z^n]F(z) \\
&= n!\frac{1}{n}[w^{n-1}] \left( \frac{1}{1-w} \right)' (ew)^n \\
&= (n-1)![w^{n-1}] \frac{e^{nw}}{(1-w)^2} \\
&= (n-1)![w^{n-1}] \sum_{k=0}^{\infty} (k+1)w^k \cdot \sum_{j=0}^{\infty} \frac{n^j}{j!} w^j \\
&= (n-1)! \sum_{j=0}^{n-1} ((n-1-j) + 1) \frac{n^j}{j!} \\
&= (n-1)! \sum_{j=0}^{n-1} (n-j) \frac{n^j}{j!} \\
&= (n-1)! \left( \sum_{j=0}^{n-1} n \frac{n^j}{j!} - \sum_{j=0}^{n-1} j \frac{n^j}{j!} \right) \\
&= (n-1)! \left( \sum_{j=0}^{n-1} \frac{n^{j+1}}{j!} - \sum_{j=1}^{n-1} \frac{n^j}{(j-1)!} \right) \\
&= (n-1)! \left( \sum_{j=0}^{n-1} \frac{n^{j+1}}{j!} - \sum_{j=0}^{n-2} \frac{n^{j+1}}{j!} \right) \\
&= (n-1)! \frac{n^n}{(n-1)!} \\
&= n^n
\end{aligned}$$

Now consider  $K = \log(1 - T)^{-1}$ . Then

$$K_n = n^{n-1}Q(n)$$

where

$$Q_n = 1 + \frac{n-1}{n} + \frac{(n-1)(n-2)}{n^2} + \dots$$

From Note II.18 we have

$$n![z^n] \frac{T(z)^k}{k!} = \binom{n-1}{k-1} n^{n-k} \implies [z^n] \frac{T(z)^k}{k} = (k-1)! \binom{n-1}{k-1} n^{n-k} = \frac{(n-1)!}{(n-k)!} n^{n-k}$$

Consequently,

$$\begin{aligned}
 K_n &= n![z^n]K(z) \\
 &= n![z^n] \sum_{j=1}^{\infty} \frac{T(z)^j}{j} \\
 &= \sum_{j=1}^{\infty} n![z^n] \frac{T(z)^j}{j} \\
 &= \sum_{j=1}^{\infty} \frac{(n-1)!}{(n-j)!} n^{n-j} \\
 &= n^{n-1} \sum_{j=1}^{\infty} \frac{(n-1)!}{(n-j)! n^{j-1}} \\
 &= n^{n-1} \left( 1 + \frac{n-1}{n} + \frac{(n-1)(n-2)}{n^2} + \dots \right)
 \end{aligned}$$

## 2.6 Additional constructions

## 2.7 Perspective

# Chapter 3

## COMBINATORIAL PARAMETERS AND MULTIVARIATE GENERATING FUNCTIONS

Content for chapter 3 goes here...

- 3.1 An introduction to bivariate generating functions (BGFs)
- 3.2 Bivariate generating functions and probability distributions
- 3.3 Inherited parameters and ordinary MGFs
- 3.4 Inherited parameters and exponential MGFs
- 3.5 Recursive parameters
- 3.6 Complete generating functions and discrete models
- 3.7 Additional constructions
- 3.8 Extremal parameters
- 3.9 Perspective